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# Student Performance Prediction System

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**Project Id: AIML-7**

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# 1. Problem Description

Type-2 Diabetes is one of the most common chronic diseases worldwide. It occurs when the body becomes resistant to insulin or does not produce enough insulin to maintain normal blood glucose levels. Early diagnosis and prediction are important because untreated diabetes can lead to severe complications such as kidney failure, heart disease, nerve damage, and vision problems.

Machine Learning techniques can assist healthcare professionals by analyzing medical data and identifying patients who are at high risk of developing diabetes. This project uses Logistic Regression along with preprocessing techniques to build a predictive system for Type-2 Diabetes risk assessment.

The project focuses on building a clinical risk assessment system using synthetic healthcare data. The dataset contains medical features such as Age, BMI, Glucose Level, Insulin Level, Blood Pressure, and HbA1c. The aim is to predict whether a patient is diabetic or non-diabetic using machine learning classification algorithms.

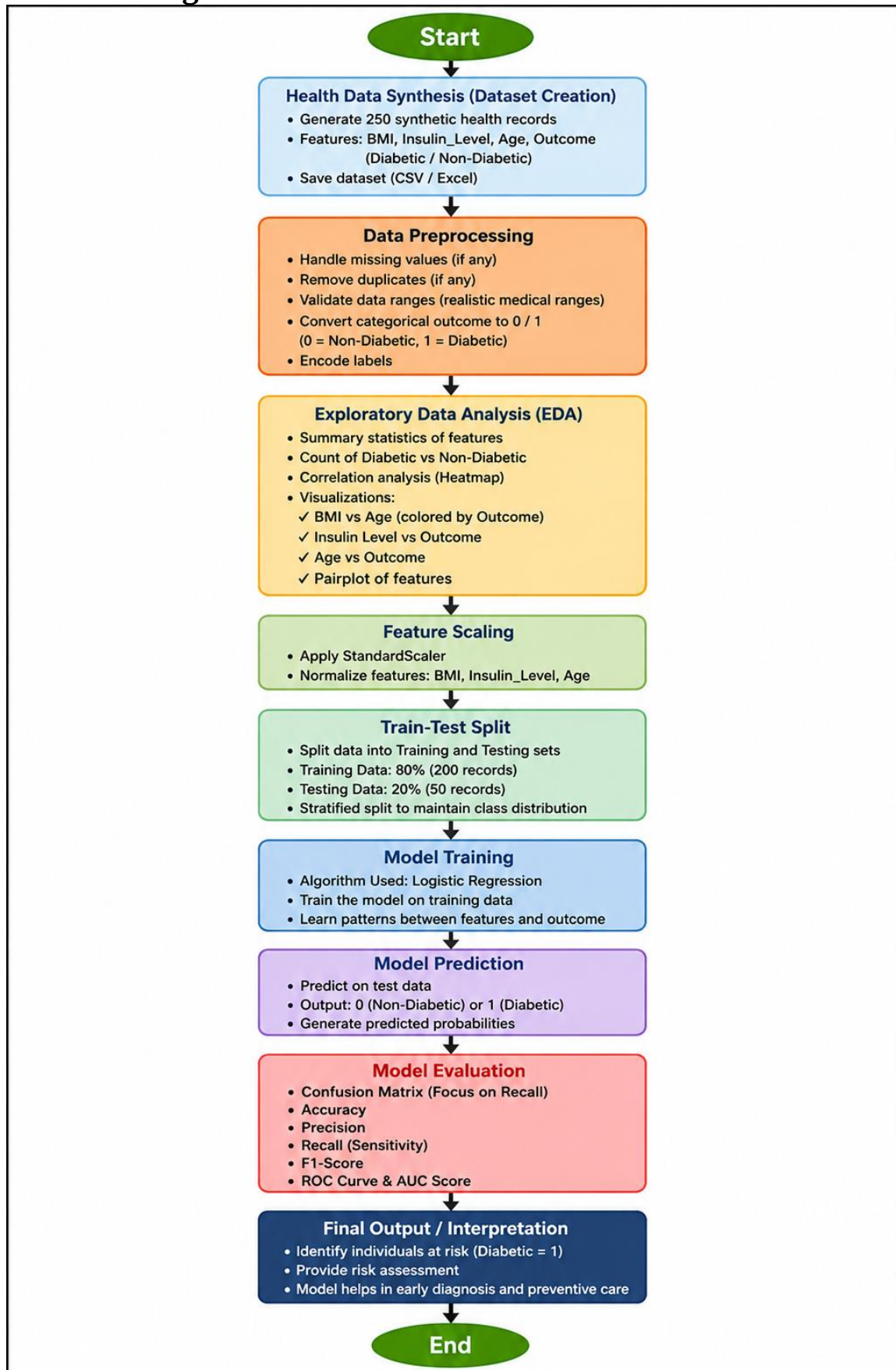
## 1.1 Dataset Description

The cleaned dataset contains 3381 patient records with 13 attributes. The dataset includes:

- Patient\_ID
- Age
- Gender
- BMI
- Glucose\_Level
- Insulin
- Blood\_Pressure
- Skin\_Thickness
- Diabetes\_Pedigree\_Function
- HbA1c
- Cholesterol
- Medication
- Outcome

The raw dataset contained inconsistent and missing values such as text entries in numerical columns and missing BMI values. These issues were corrected during preprocessing.

## 1.2 Flow Diagram



## 2. Modules for Project Implementation

### 2.1 Modules Used:

- **NumPy** – Used for numerical computations and array operations.
- **Pandas** – Used for dataset loading, cleaning, preprocessing, and manipulation.
- **Matplotlib** – Used for creating graphs and visualizations such as scatter plots.
- **Seaborn** – Used for advanced visualizations and confusion matrix heatmaps.
- **Scikit-learn** – Used for machine learning algorithms, preprocessing, and evaluation metrics.
- **StandardScaler** – Used to normalize numerical features for better model performance.
- **LabelEncoder** – Used to convert categorical data into numerical form.
- **Train\_Test\_Split** – Used to divide the dataset into training and testing sets.
- **LogisticRegression** – Used to classify patients as diabetic or non-diabetic.
- **Confusion Matrix** – Used to evaluate prediction performance of the model.
- **Accuracy Score** – Used to measure overall prediction accuracy.
- **Precision Score** – Used to measure correctness of predicted diabetic cases.
- **Recall Score** – Used to measure the model's ability to identify actual diabetic patients.

## 2.2 PlatformUsed:

- **Jupyter Notebook** – Used for writing, executing, and testing Python code along with data visualization and machine learning implementation.
- **MS Excel** – Used for storing, viewing, and manually analyzing the healthcare datasets.
- **Python** – Used as the programming language for data preprocessing, machine learning, and visualization tasks.

### **3. Machine Learning Model/ Searching Technique Used:**

#### **Logistic Regression:**

Logistic Regression is a supervised machine learning classification algorithm used for binary classification problems. In this project, it is used to predict whether a patient is diabetic or non-diabetic based on healthcare parameters such as BMI, Age, Glucose Level, and Insulin Level.

The model calculates the probability of diabetes occurrence using input features and classifies the output into two categories:

- 0 → Non-Diabetic
- 1 → Diabetic

#### **Why Logistic Regression was Used:**

- Suitable for binary classification problems.
- Simple and efficient algorithm.
- Provides fast training and prediction.
- Easy to interpret and analyze results.
- Works effectively with healthcare prediction systems.

#### **Working of Logistic Regression:**

1. The cleaned and scaled healthcare dataset is provided as input.
2. The model learns relationships between features and diabetes outcome during training.
3. The trained model predicts diabetes risk on testing data.
4. The prediction results are evaluated using Accuracy, Precision, Recall, and Confusion Matrix.

#### **Role in the Project:**

The Logistic Regression model helps in identifying individuals at risk of Type-2 Diabetes at an early stage, which supports preventive healthcare and clinical risk assessment.

## 4. Evaluation metrics used :

### 1. Accuracy:

Accuracy measures the overall correctness of the machine learning model. It shows how many predictions were correctly classified out of the total predictions made by the model.

Formula:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

Where:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

Purpose in Project: Accuracy was used to measure the overall performance of the diabetes prediction model.

### 2. Precision:

Precision measures how many patients predicted as diabetic were actually diabetic.

Formula:

$$\text{Precision} = \frac{TP}{TP+FP}$$

Purpose in Project:

Precision was used to reduce false positive predictions in diabetes detection.

### 3. Recall:

Recall measures the ability of the model to correctly identify actual diabetic patients.

Formula:

$$\text{Recall} = \frac{TP}{TP+FN}$$

Purpose in Project: Recall is very important in healthcare applications because missing an actual diabetic patient can lead to serious health complications.

### 4. Confusion Matrix:

Confusion Matrix is a performance evaluation table used to compare actual values with predicted values.

Structure of Confusion Matrix

| Actual / Predicted | Positive            | Negative            |
|--------------------|---------------------|---------------------|
| Positive           | True Positive (TP)  | False Negative (FN) |
| Negative           | False Positive (FP) | True Negative (TN)  |

Purpose in Project: The confusion matrix was used to analyze correct and incorrect predictions made by the Logistic Regression model.

### **5. True Positive Rate (TPR):**

True Positive Rate measures the proportion of actual diabetic patients correctly identified by the model.

Formula:

$$\text{TPR} = \text{TP} / (\text{TP} + \text{FN})$$

Purpose in Project: TPR was used to evaluate how effectively the model detects diabetic cases.

### **6. False Positive Rate (FPR):**

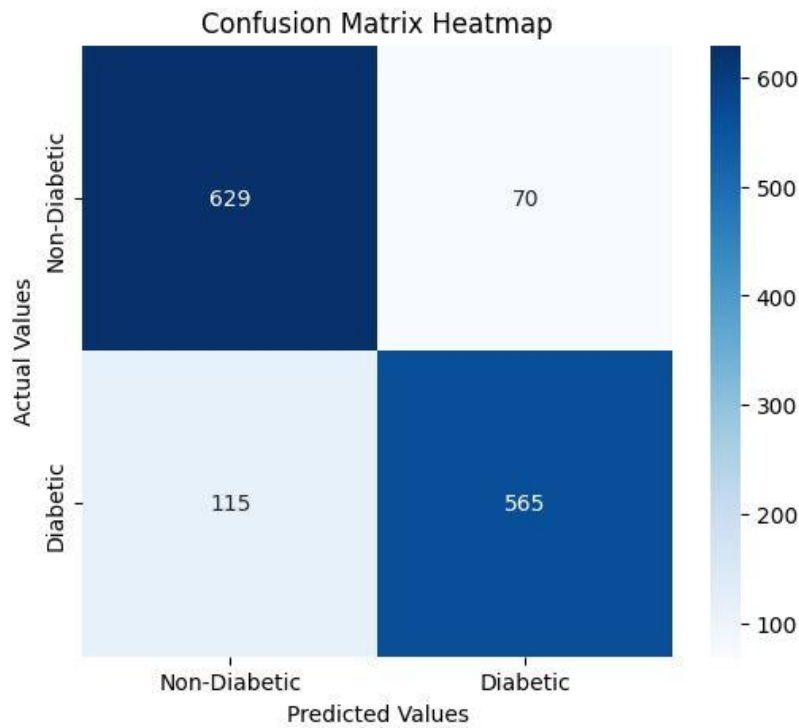
False Positive Rate measures the proportion of non-diabetic patients incorrectly classified as diabetic.

Formula:

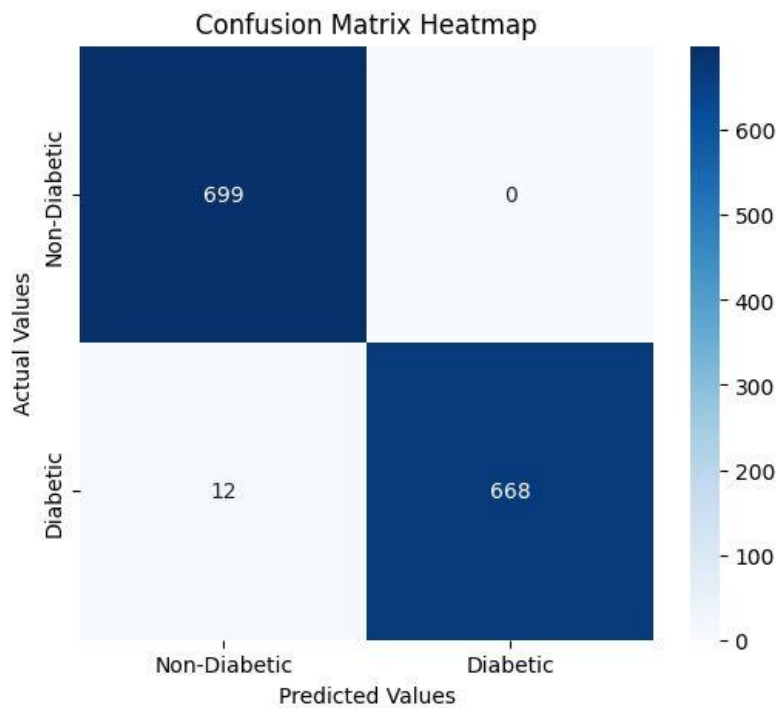
$$\text{FPR} = \text{FP} / (\text{FP} + \text{TN})$$

Purpose in Project: FPR was used to measure incorrect positive predictions by the model.

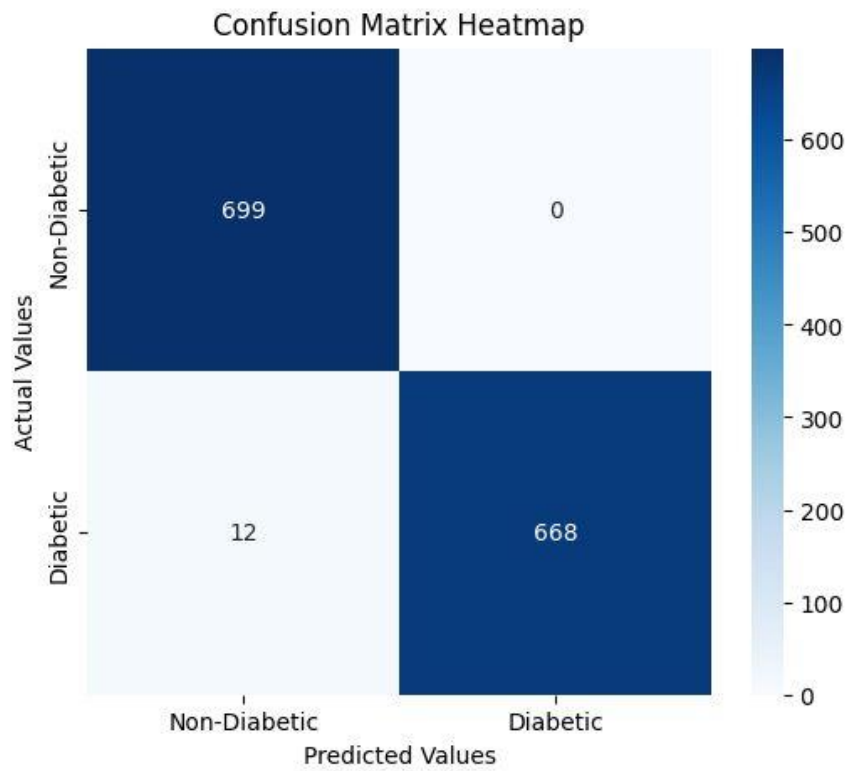
## 5. Results and Visualizations:



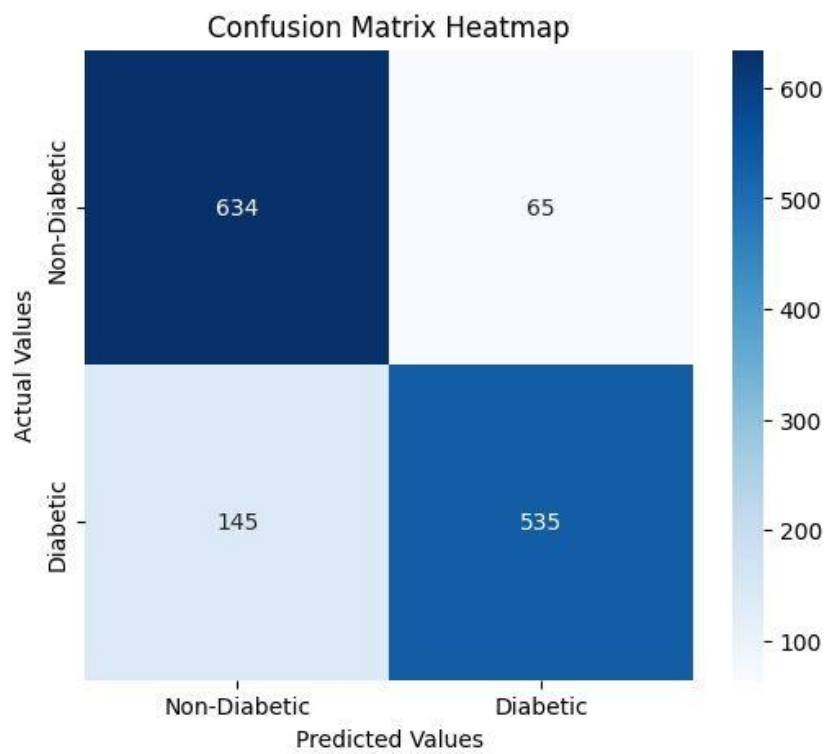
### A) LOGISTIC REGRESSION



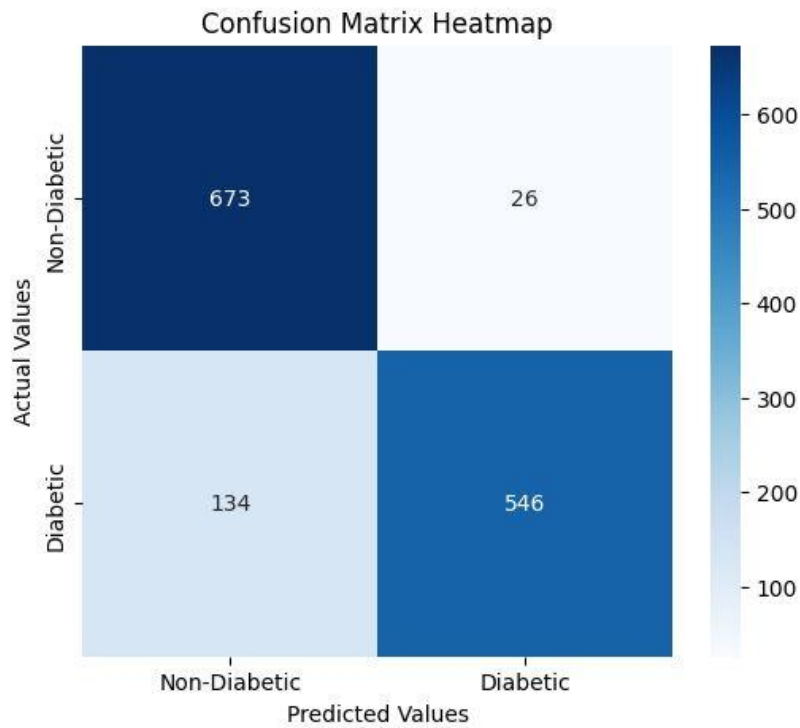
### B) RANDOM FOREST



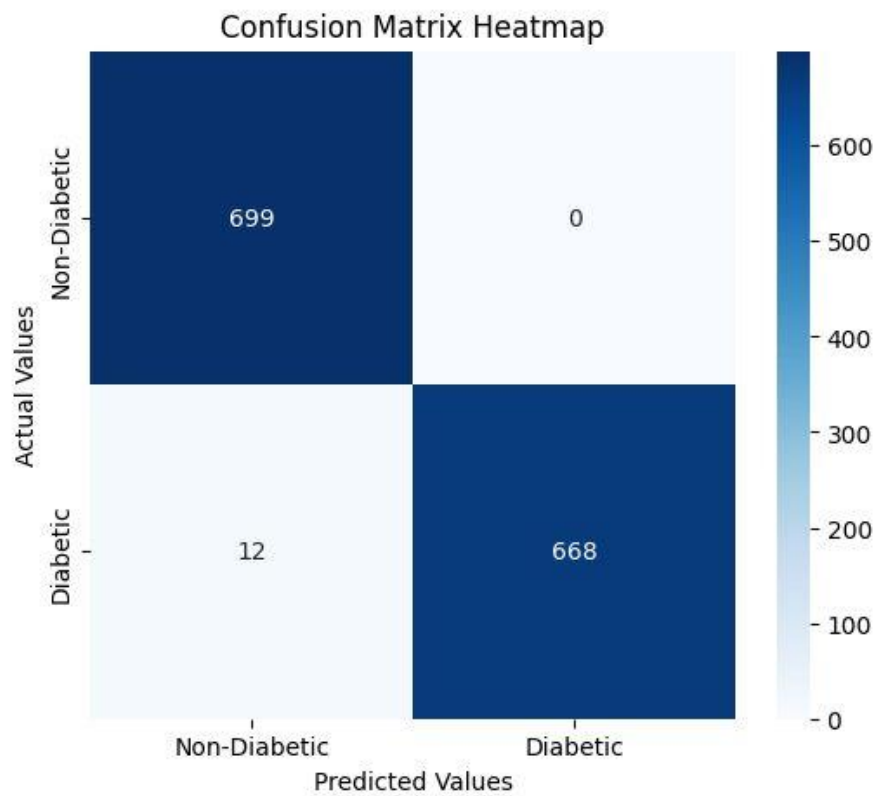
### C) KMEANS



### D) LINEAR REGRESSION



**E) KNN**



**F) DECISION TREE**

## A)LOGISTIC REGRESSION

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.85      | 0.90   | 0.87     | 699     |
| 1            | 0.89      | 0.83   | 0.86     | 680     |
| accuracy     |           |        | 0.87     | 1379    |
| macro avg    | 0.87      | 0.87   | 0.87     | 1379    |
| weighted avg | 0.87      | 0.87   | 0.87     | 1379    |

## B)RANDOM FOREST

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| Not Diabetic | 0.98      | 1.00   | 0.99     | 699     |
| Diabetic     | 1.00      | 0.98   | 0.99     | 680     |
| accuracy     |           |        | 0.99     | 1379    |
| macro avg    | 0.99      | 0.99   | 0.99     | 1379    |
| weighted avg | 0.99      | 0.99   | 0.99     | 1379    |

## C)KMEANS

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| Not Diabetic | 0.51      | 0.58   | 0.54     | 3447    |
| Diabetic     | 0.51      | 0.43   | 0.47     | 3447    |
| accuracy     |           |        | 0.51     | 6894    |
| macro avg    | 0.51      | 0.51   | 0.50     | 6894    |
| weighted avg | 0.51      | 0.51   | 0.50     | 6894    |

## D) LINEAR REGRESSION

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| Not Diabetic | 0.81      | 0.91   | 0.86     | 699     |
| Diabetic     | 0.89      | 0.79   | 0.84     | 680     |
| accuracy     |           |        | 0.85     | 1379    |
| macro avg    | 0.85      | 0.85   | 0.85     | 1379    |
| weighted avg | 0.85      | 0.85   | 0.85     | 1379    |

## E)KNN

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| Not Diabetic | 0.83      | 0.96   | 0.89     | 699     |
| Diabetic     | 0.95      | 0.80   | 0.87     | 680     |
| accuracy     |           |        | 0.88     | 1379    |
| macro avg    | 0.89      | 0.88   | 0.88     | 1379    |
| weighted avg | 0.89      | 0.88   | 0.88     | 1379    |

## F)DECISION TREE

Classification Report:

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| Not Diabetic | 0.98      | 1.00   | 0.99     | 699     |
| Diabetic     | 1.00      | 0.98   | 0.99     | 680     |
| accuracy     |           |        | 0.99     | 1379    |
| macro avg    | 0.99      | 0.99   | 0.99     | 1379    |
| weighted avg | 0.99      | 0.99   | 0.99     | 1379    |

## **6. Conclusion and Future Scope:**

The project successfully demonstrates the use of machine learning for Clinical Risk Assessment of Type-2 Diabetes. A Logistic Regression model was developed using healthcare parameters such as Age, BMI, Glucose Level, and Insulin Level to predict whether a patient is diabetic or non-diabetic.

Various preprocessing techniques such as data cleaning, label encoding, feature scaling, and dataset balancing were applied to improve the quality of the dataset and model performance. The model was evaluated using Accuracy, Precision, Recall, and Confusion Matrix, where recall played an important role in identifying actual diabetic patients correctly.

The results and visualizations show that machine learning can effectively assist healthcare professionals in early diabetes prediction and preventive healthcare systems.

### **Future Scope:**

- The project can be improved using larger real-world healthcare datasets for better prediction accuracy.
- Advanced machine learning algorithms such as Random Forest, Decision Tree, Support Vector Machine, and Neural Networks can be implemented for comparison and improved performance.
- A web-based or mobile healthcare application can be developed for real-time diabetes risk prediction.
- Real-time patient monitoring systems and IoT devices can be integrated with the prediction model.
- Cloud-based healthcare analytics can be used for large-scale medical data processing and storage.
- Additional healthcare parameters and medical history data can be included to make the prediction system more accurate and reliable.

## 7. References:

Books:

- LetUsPython–YashavantKanetkar
- Online Resources:
  1. GeeksforGeeks–<https://www.geeksforgeeks.org/>
  2. TutorialsPoint–<https://www.tutorialspoint.com/cprogramming/index.htm>
  3. StackOverflow–<https://stackoverflow.com>